

```
def patnumr():  
    f=open('esther pat.txt','w+')  
    f.write('PAT NUM R\n-----\n')  
    for i in range(1,6):  
        for j in range(6,i,-1):  
            f.write(' ')  
        for k in range(0,i):  
            f.write(str(i)+' ')  
        f.write('\n')  
    f.close()
```

patnumr()

```
def patnumc():  
    f=open('esther pat.txt','a+')  
    f.write('PAT NUM C\n-----\n')  
    for i in range(1,6):  
        for j in range(6,i,-1):  
            f.write(' ')  
        for k in range(0,i):  
            f.write(str(k)+' ')  
        f.write('\n')  
    f.close()
```

patnumc()

```
patstar()
```

```
def patstar():  
    f=open('esther pat.txt','a+')  
    f.write('RAT STAR \n-----\n')  
    for i in range(1,6):  
        for j in range(6,i,-1):  
            f.write(' ')  
        for k in range(0,i):  
            f.write(str('*')+' ')  
        f.write('\n')  
    f.close()
```

```
patstar()
```

```
def patupperr():  
    f=open('esther pat.txt','a+')  
    f.write('PAT UPPER R\n-----\n')  
    for i in range(0,5):  
        for j in range(5,i,-1):  
            f.write(' ')  
        for k in range(0,i):  
            f.write(str(chr(i+64)+' '))  
        f.write('\n')  
    f.close()
```

```
patupperr()
```

```
def patupperc():  
    f=open('esther pat.txt','a+')  
    f.write('PAT UPPER C\n-----\n')  
    for i in range(1,5):  
        for j in range(5,i,-1):  
            f.write(' ')  
        for k in range(0,i):  
            f.write(str(chr(k+65)+' '))  
        f.write('\n')  
    f.close()
```

patupperc()

```
def patlowerr():  
    f=open('esther pat.txt','a+')  
    f.write('PAT LOWER R\n-----\n')  
    for i in range(1,5):  
        for j in range(5,i,-1):  
            f.write(' ')  
        for k in range(0,i):  
            f.write(str(chr(i+96)+' '))  
        f.write('\n')  
    f.close()
```

patlowerr()

```
def patlowerc():  
    f=open('esther pat.txt', 'a+')  
    f.write('PAT LOWER C\n-----\n')  
    for i in range(1,5):  
        for j in range(5,i,-1):  
            f.write(' ')  
        for k in range(0,i):  
            f.write(str(chr(k+97)+' '))  
        f.write('\n')  
    f.close()
```

patlowerc()

```
def patnumri():  
    f=open('esther pat.txt', 'a+')  
    f.write('PAT NUM INVERSE R\n-----\n')  
    for i in range(6,-1,-1):  
        for j in range(6,i,-1):  
            f.write(' ')  
        for k in range(0,i):  
            f.write(str(i)+' ')  
        f.write('\n')  
    f.close()
```

patnumri()

```
def patnumci():  
    f=open('esther pat.txt','a+')  
    f.write('PAT NUM INVERSE C\n-----\n')  
    for i in range(6,-1,-1):  
        for j in range(6,i,-1):  
            f.write(' ')  
        for k in range(0,i):  
            f.write(str(k)+' ')  
        f.write('\n')  
    f.close()
```

patnumci()

```
def patstari():  
    f=open('esther pat.txt','a+')  
    f.write('PAT STAR INVERSE \n-----\n')  
    for i in range(6,-1,-1):  
        for j in range(6,i,-1):  
            f.write(' ')  
        for k in range(0,i):  
            f.write(str('*')+' ')  
        f.write('\n')  
    f.close()
```

patstari()

PAT.py

```
def patupperri():  
    f=open('esther pat.txt','a+')  
    f.write('PAT UPPER INVERSE R\n-----\n')  
    for i in range(5,-1,-1):  
        for j in range(5,i,-1):  
            f.write(' ')  
        for k in range(0,i):  
            f.write(str(chr(i+64))+ ' ')  
        f.write('\n')  
    f.close()
```

patupperri()

```
def patupperci():  
    f=open('esther pat.txt','a+')  
    f.write('PAT UPPER INVERSE C\n-----\n')  
    for i in range(5,-1,-1):  
        for j in range(5,i,-1):  
            f.write(' ')  
        for k in range(0,i):  
            f.write(str(chr(k+65))+ ' ')  
        f.write('\n')  
    f.close()
```

patupperci()

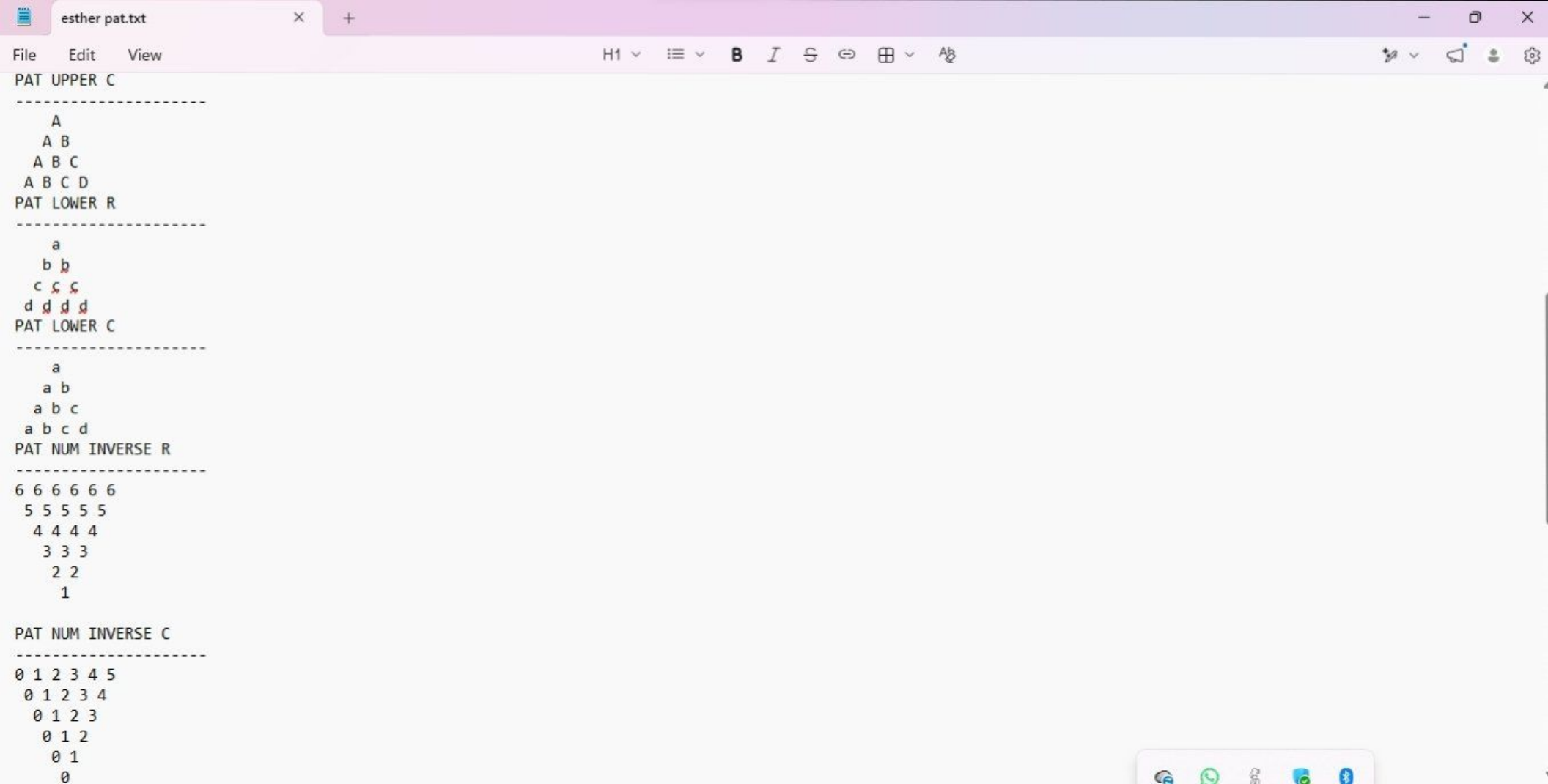

```
def patupper():  
    f=open('esther pat.txt','a+')  
    f.write('PAT UPPER INVERSE C\n-----\n')  
    for i in range(5,-1,-1):  
        for j in range(5,i,-1):  
            f.write(' ')  
        for k in range(0,i):  
            f.write(str(chr(k+65))+ ' ')  
        f.write('\n')  
    f.close()
```

patupperci()

```
def patlowerri():  
    f=open('esther pat.txt','a+')  
    f.write('PAT LOWER INVERSE R\n-----\n')  
    for i in range(5,-1,-1):  
        for j in range(5,i,-1):  
            f.write(" ")  
        for k in range(0,i):  
            f.write(str(chr(i+96))+ ' ')  
        f.write('\n')  
    f.close()
```

patlowerri()

```
File Edit View H1 [Menu] B I S [Menu] [Menu] [Menu]
PAT NUM R
-----
  1
 2 2
3 3 3
4 4 4 4
5 5 5 5 5
PAT NUM C
-----
  0
 0 1
0 1 2
0 1 2 3
0 1 2 3 4
RAT STAR
-----
  *
 * *
* * *
* * * *
* * * * *
PAT UPPER R
-----
 |
A
B B
C C C
D D D D
PAT UPPER C
```

PAT UPPER C

A
A B
A B C
A B C D
PAT LOWER R

a
b b
c c c
d d d d
PAT LOWER C

a
a b
a b c
a b c d
PAT NUM INVERSE R

6 6 6 6 6 6
5 5 5 5 5
4 4 4 4
3 3 3
2 2
1

PAT NUM INVERSE C

0 1 2 3 4 5
0 1 2 3 4
0 1 2 3
0 1 2
0 1
0

PAT STAR INVERSE

```
*****
* * * * *
* * * * *
* * * *
* * *
* *
* *
```

PAT UPPER INVERSE R

```
E E E E E
D D D D
C C C
B B
A
```

PAT UPPER INVERSE C

```
A B C D E
A B C D
A B C
A B
A
```

PAT LOWER INVERSE R

```
e e e e e
d d d d
c c c
b b
a
```

